

Ellen Poulsen

 Ellen Poulsen |  ellenpoulsen5@gmail.com |  +1 (541) 654-2129

EDUCATION

2025 — 2029 **B.S. Aerospace Engineering Degree Candidate** GPA: 3.66/4.0
California Polytechnic State University, San Luis Obispo

2023 **Oregon State Seal of Biliteracy** — Spanish

WORK EXPERIENCE

Test Engineering Intern — Empirical Systems Aerospace *June 2026 — Aug. 2026*
San Luis Obispo, CA

- Developed servo testing procedures and software to analyze frequency response, working in parallel with mechanical design team on servo test stand to accurately measure angular velocity and torque. Preliminary design review performed and product considered for application in both research and development as well as production line testing.
- Designed Python based data acquisition software to measure phase delay and bandwidth, with focus on graphical data representation and user interface for broad usage requirements. Operated with both CAN bus and PWM communication protocols for use across different vehicle applications and systems.
- Calibrated data acquisition timing across multiple testing devices to ensure precise sensor telemetry collection, allowing analysis for high speed performance metrics required by flight vehicles.
- Managed project version control and code collaboration using Python, VS Code, and GitLab, alongside thorough documentation of product development for project handoff and future development after internship completion.

Observatory Assistant — Cal Poly Physics Department *May 2026 — Present*

- Operating 14-inch Meade LX600 ACF Telescope for public viewing events.
- Leading educational observing sessions for student groups, presenting key astrophysics concepts and managing night-sky targets.

PROJECTS

Cal Poly Space Systems — Liquids Project

- Development of liquid rocket engines including co-leading work on current regenerative engine project: developing testing dates and metrics, task delegation, and goals for future vehicle integration.
- Performing high pressure fluids testing safely through continuously updated procedures and familiarization with modes of failure.
- Improving and maintaining test stand by ensuring proper operation and reading from pressure transducers, cleaning lines, and testing proper operation of pneumatic and electrical systems.

LEADERSHIP EXPERIENCE

Cal Poly Space Systems Lead — SILLE (Student Introductory Liquids Learning Engine)

- Teaching students how to design and manufacture their own small liquid rocket engine, present and receive feedback from design reviews, and conduct safe testing on their engines up to 50 lbf thrust.
- Organization of over 40 students into teams, teaching basics of liquids rocketry, foundational understanding of isentropic equations and associated variables, machining, and safe testing practices.
- Focus on member retention and transfer of knowledge through dedicated mentorship, ensuring students feel supported and encouraging their continued involvement to promote longevity and growth of the liquids team.
- Public speaking and academic presentation to large audiences alongside proficiency in group organization and communication utilizing Microsoft Teams, PowerPoint, Forms, and Excel.

Society of Women Engineers at Cal Poly

- Advocating for and supporting women underrepresented in engineering fields through community outreach, providing educational opportunities aimed at advancing access to STEM and inspiring the next generation of women engineers.